Sanket Desai

DevOps / Cloud Engineer | Backend Developer | Automation & Infrastructure
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Professional Summary

B.Tech Computer Science graduate aspiring to build a career in DevOps and cloud engineering, with hands-on experience in backend development, containerisation, databases, and system automation. I enjoy working at the intersection of development and infrastructure, building applications, containerising services, configuring environments, and understanding how systems run reliably in production. I have worked with Docker, Git, Linux, Python, Java, Spring Boot, FastAPI, PostgreSQL, and related development tools. I learn best by building, troubleshooting, and improving real systems.

Technical Skills

Languages: Python, Java, Kotlin, SQL, JavaScript, TypeScript
Backend/APIs: FastAPI, REST APIs
Data/Messaging: PostgreSQL, Room
AI/ML: LLMs, RAG, Ollama, vector search, GNNs, agentic workflows
UI/UX: React, Jetpack Compose, Android, HTML, CSS, responsive design, UI/UX principles
Infrastructure/DevOps: Docker, Docker Compose, CI/CD, Linux, Git, GitHub
Concepts: Idempotency, transactional outbox, dead-letter queues, exponential backoff, eventual consistency, observability, DSA, OS, networking, DBMS
Tools: Git, GitHub, Scapy, NumPy

Projects

Resilient Order Processing System *Kafka, Spring Boot, PostgreSQL, Docker* *View Project ↗*

- Improved order-processing reliability by ensuring duplicate requests, transient payment failures, and message-delivery failures can be handled safely without losing or duplicating orders.
- Designed an event-driven order pipeline with persistent idempotency keys (UUIDv7) to guarantee safe processing under at-least-once message delivery.
- Implemented a polling transactional outbox with exponential backoff and dead-letter queue routing to reliably publish events and recover failed messages.

Uptime Monitoring System *Python, FastAPI, PostgreSQL, React, Docker Compose, Grafana* *View Project ↗*

- Improved service reliability and operational visibility by continuously detecting endpoint failures, tracking incidents, and delivering automated alerts before issues remain unnoticed.
- Built an asynchronous monitoring engine that schedules HTTP health checks and persists service and incident states in PostgreSQL.
- Implemented email and webhook alerting with Grafana dashboards to provide centralized visibility into service health and failures.

AI Knowledge System (RAG) *Python, FastAPI, LLMs, Ollama, ChromaDB* *View Project ↗*

- Enabled reliable, privacy-preserving access to internal documentation by generating cited answers from company knowledge while keeping sensitive documents and inference within the local environment.
- Built a RAG pipeline using hybrid vector and keyword retrieval with cross-encoder re-ranking to improve contextual relevance.
- Ran embeddings and LLM inference locally through Ollama, eliminating the need to send company documents to external AI services.

Agentic Security Operations Center *Python, Scapy, NumPy, Docker* *View Project ↗*

- Automated network-threat detection and response by continuously analyzing traffic, identifying anomalous behavior, and triggering firewall-based mitigation actions.
- Built a containerized monitoring pipeline that captures network flows, generates graph representations, and evaluates traffic anomalies.
- Automated Linux firewall responses using LangGraph-style agent orchestration for deterministic security workflows.

Text-to-SQL Engine *Python, FastAPI, SQL, Ollama* *View Project ↗*

- Reduced the complexity of querying relational databases by allowing users to retrieve structured data through natural-language questions while validating generated SQL before execution.
- Built a FastAPI service that converts natural-language questions into executable SQL using a locally hosted LLM.
- Added AST-based validation to detect unsafe, malformed, or unsupported queries before database execution.

ForgeGym Workout Tracker *Kotlin, Android, Jetpack Compose, Room* *View Project ↗*

- Simplified workout tracking with a fully offline mobile application for creating routines, recording training sessions, and reviewing workout history without a backend service.
- Developed an Android application using Kotlin and Jetpack Compose with a declarative, component-based UI architecture.

Certifications

JPMorgan Chase Software Engineering Virtual Experience Program — Forage *2026*
JPMorgan Chase Quantitative Research Virtual Experience Program — Forage *2026*
Electronic Arts Software Engineering Virtual Experience Program — Forage *2026*
CTF Participation Certificate — MIT-WPU *2026*

Education

B.Tech, Computer Science and Engineering *2023 – 2026*
Deccan Education Society Pune University CGPA: 7.05
Coursework: Data Structures and Algorithms, Operating Systems, Computer Networks, DBMS, Software Engineering

Diploma, Electronics and Telecommunication *2020 – 2023*
Deccan Education Society Pune University Percentage: 77